A Growth–Statics Model for Paediatric Prosthetic Socket Replacement

SCUDEM XI 2026, Problem A — full derivation

Working derivation

November 4, 2026

Abstract

We build a mathematical model that predicts *whether, and at what age*, a child's transradial (below-elbow) prosthetic socket reaches a point of substantial change that warrants adjustment or replacement as the residual limb grows. The model has three layers: (i) a **growth ordinary differential equation** (ODE) for residual-limb length and circumference; (ii) a **socket-fit map** turning the growth trajectory into tightness and distal-clearance signals; and (iii) a **planar lift statics** block covering a small *set* of actions (a hold and a curl, plus a straight-arm reach used only in the optional shoulder-load extension), whose axial load feeds back onto the clearance criterion so that the lifted load changes the predicted threshold. A closed-form **threshold** rule combines three criteria — tightness, distal pistoning, and prosthetic length discrepancy — to return the time-to-intervention T^*, and a **sensitivity analysis** reports T^* against fitting age with bands from the uncertain tolerances. The prediction is validated by fitting the model to lower-leg data (where paediatric casefiles exist) before predicting on the arm. All intermediate algebra is shown.

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1 Notation and type key

Symbol	Meaning	Unit	Type
a	age (independent variable)	yr	independent
$L(a)$	residual-limb length	cm	state (ODE)
$C(a)$	residual-limb circumference	cm	state (ODE)
r_L, r_C	length / circumference growth rates	yr^{-1}	parameter
K_L, K_C	asymptotic (adult) length / circumference	cm	parameter
ρ	residual-to-intact length ratio	–	parameter
$M_b(a)$	body mass	kg	derived
a_0	age at socket fitting	yr	input
s_C	intentional circumferential oversize	–	input (swept)
δ_0	initial distal clearance	cm	input (swept)
C_s, D_s	socket inner circumference / depth	cm	derived
$\varepsilon_c(a)$	tightness $(C - C_s)/C_s$	–	derived
$\delta(a)$	distal clearance $D_s - L$	cm	derived
$\varepsilon_{\max}$	tolerable tightness	–	parameter (swept)
m_p, m_o	prosthesis / object mass	kg	input
ℓ_p, ℓ_h	elbow→prosthesis-CoM / elbow→hand arms (fixed at a_0)	m	input
$\ell_{\mathrm{fa}}(a)$	intact forearm length (Otmani #11 / NHANES), grows	m	parameter
$\ell_{\mathrm{ua}}(a)$	upper-arm length (grows; extension only)	m	parameter
K_Λ, r_Λ	asymptote / rate of intact-forearm growth	m, yr^{-1}	parameter
$\Delta L_{\max}$	tolerable prosthetic length discrepancy	m	parameter (swept)
φ	forearm angle from vertical (90° = horizontal)	deg	input
M_s, F_{ax}	socket bending moment / axial load	N·m, N	derived
k_s	liner+soft-tissue axial stiffness	N/cm	parameter (swept)
u_{lift}	curl pistoning displacement $\lvert F_{\mathrm{ax}}\rvert/k_s$	cm	derived
a_c^*, a_d^*, a_ℓ^*	tightness / clearance / length-discrepancy ages	yr	output
a^*, T^*	threshold age / time-to-intervention	yr, mo	output

Table 1: Variables. Type key: state = solved by the ODE; parameter = fitted or from literature; input = chosen (swept where noted); derived = computed; output = headline result. Clamp notation $\langle z \rangle_+ = \max(z, 0)$.

2 Layer 1 — the growth ODE

2.1 Model statement

We treat residual-limb length L and circumference C as logistic growth processes decelerating toward adult asymptotes K_L, K_C:

$$\frac{dL}{da} = r_L\, L\left(1 - \frac{L}{K_L}\right), \qquad \frac{dC}{da} = r_C\, C\left(1 - \frac{C}{K_C}\right). \tag{1}$$

Collecting the state $x(a) = \big[L(a),\, C(a)\big]^\top$ and parameters $\theta = (r_L, K_L, r_C, K_C, \rho)$,

$$\frac{dx}{da} = f(x;\theta), \qquad f(x;\theta) = \begin{bmatrix} r_L\, L\,(1 - L/K_L) \\ r_C\, C\,(1 - C/K_C) \end{bmatrix}. \tag{2}$$

2.2 Residual-vs-intact scaling

Because residual-limb growth is stunted relative to the intact limb, intact anthropometric data (NHANES) is mapped to the residual limb by

$$L_{\text{res}}(a) = \rho\, L_{\text{intact}}(a), \qquad 0 < \rho \le 1. \tag{3}$$

Equivalently one may fit θ directly to residual-limb data; both give the same $x(a)$.

2.3 Closed-form solution (all intermediate steps)

Equation (1) is separable. For the length equation,

$$\int \frac{dL}{L\,(1 - L/K_L)} = \int r_L\, da. \tag{4}$$

Write $L(1 - L/K_L) = \dfrac{L(K_L - L)}{K_L}$, so the integrand is $\dfrac{K_L}{L(K_L - L)}$. Partial fractions:

$$\frac{K_L}{L(K_L - L)} = \frac{A}{L} + \frac{B}{K_L - L} \;\Rightarrow\; K_L = A(K_L - L) + B\,L. \tag{5}$$

Setting $L = 0$ gives $A = 1$; setting $L = K_L$ gives $B = 1$. Hence

$$\int \left(\frac{1}{L} + \frac{1}{K_L - L}\right) dL = \int r_L\, da \;\Rightarrow\; \ln L - \ln(K_L - L) = r_L\, a + c_L. \tag{6}$$

Fixing a baseline age a_b with $L(a_b) = L_b$ gives $c_L = \ln \frac{L_b}{K_L - L_b} - r_L a_b$, so

$$\frac{L}{K_L - L} = \frac{L_b}{K_L - L_b}\, e^{\,r_L(a - a_b)} \;\Rightarrow\; \boxed{L(a) = \frac{K_L}{1 + B_L\, e^{-r_L(a - a_b)}}}, \qquad B_L = \frac{K_L - L_b}{L_b}. \tag{7}$$

By the identical argument,

$$\boxed{C(a) = \frac{K_C}{1 + B_C\, e^{-r_C(a - a_b)}}}, \qquad B_C = \frac{K_C - C_b}{C_b}, \quad C(a_b) = C_b. \tag{8}$$

These closed forms make the threshold ages analytic (Section 5).

Remark 1 (growth spurt)**.** A single logistic curve is monotone and cannot reproduce the pubertal spurt (ages 10–14). A time-varying rate $r_L(a)$ or a two-phase logistic is a drop-in replacement for f; every later layer is untouched because it uses only the *solution* $x(a)$. The spurt would add a dip to the headline curve of Section 6.

3 Layer 2 — the socket-fit map

3.1 Socket built at the fitting age

At fitting age a_0 the socket is sized to the limb as it then is, with intentional oversize s_C (prosthetists build loose and take up the slack with socks) and distal clearance δ_0:

$$C_s = (1 + s_C)\, C(a_0), \qquad D_s = L(a_0) + \delta_0. \tag{9}$$

3.2 Fit signals

$$\text{tightness:} \quad \varepsilon_c(a) = \frac{C(a) - C_s}{C_s} = \frac{C(a)}{(1 + s_C)\, C(a_0)} - 1, \tag{10}$$

$$\text{distal clearance:} \quad \delta(a) = D_s - L(a) = L(a_0) + \delta_0 - L(a). \tag{11}$$

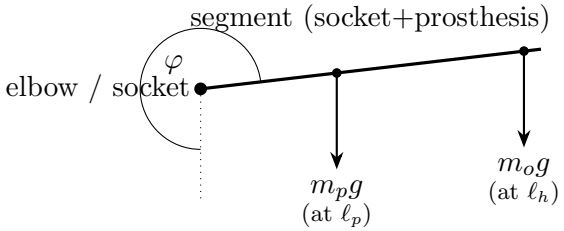
Early on $\varepsilon_c < 0$ (loose, sock-filled) and $\delta > 0$; as the limb grows, ε_c rises and δ shrinks.

4 Layer 3 — planar lift statics (a set of actions)

The problem asks for a *set* of actions and for how the interface changes *as the child grows*. We therefore (i) model two actions, (ii) place the prosthesis mass and the held object at their own *fixed* moment arms, and (iii) feed the curl's axial load back onto the clearance criterion so the lifted load changes the answer. The moment arms are fixed at the fitting age — the prosthesis is built once and does not grow — while the growth-driven length effect enters separately as the length-discrepancy criterion (16).

4.1 Free-body setup

Model the residual forearm, socket, and prosthesis as one rigid segment hinged at the elbow, in the sagittal plane. Let φ be the forearm angle from the downward vertical ($\varphi = 90°$ horizontal). Two gravitational loads act: the prosthesis weight $m_p g$ at its centre of mass (nearer the elbow) and the object weight $m_o g$ at the terminal device (the hand).



4.2 Projection with separate moment arms

Decompose each weight into components perpendicular and parallel to the segment. The perpendicular parts act at their own moment arms, so the socket bending moment sums the two contributions; the parallel parts give the axial load (sign: + distraction/pull-off, − compression/push-on):

$$M_s = g\left(m_p\, \ell_p + m_o\, \ell_h\right) \sin\varphi, \qquad F_{\text{ax}} = (m_p + m_o)\, g \cos\varphi. \tag{12}$$

Here ℓ_p (elbow to prosthesis centre of mass) and ℓ_h (elbow to hand) are *fixed at the fitting age*: the prosthesis is built once and does not grow, so M_s is constant over the socket's service life. In particular $\ell_h = \ell_{\text{fa}}(a_0) + c_h$, with ℓ_{fa} the forearm length (Otmani #11) and c_h the hand offset. The growth-driven length effect is captured separately by the length-discrepancy criterion (16).

4.3 Two actions

Two representative actions span the loading range:

Action	φ	Bending M_s	Axial F_{ax}
A: hold (max bending)	90°	$g\,(m_p\ell_p + m_o\ell_h)$	0
B: curl, hand-to-mouth	$\approx 150^\circ$	$\frac{1}{2}g\,(m_p\ell_p + m_o\ell_h)$	$-\frac{\sqrt{3}}{2}(m_p + m_o)g$ (push-on)

Action A isolates peak bending; action B produces a compressive axial load that pistons the socket onto the limb.

4.4 Pistoning displacement from the curl

In action B the compressive axial load drives the socket a distance u_{lift} onto the limb, resisted by the liner and soft-tissue axial stiffness k_s:

$$\boxed{u_{\text{lift}} = \frac{|F_{\text{ax}}|}{k_s} = \frac{(m_p + m_o)\,g\,|\cos\varphi_B|}{k_s}} \qquad (\varphi_B \approx 150^\circ \Rightarrow |\cos\varphi_B| = \tfrac{\sqrt{3}}{2}). \tag{13}$$

A heavier object or prosthesis raises u_{lift}. One new parameter, k_s, taken from literature and swept.

Remark 2 (the lift now binds)**.** With (13), the lifted load enters the clearance criterion (15) below, so increasing m_o or m_p brings the threshold earlier — the lift is no longer decoration. It also sharpens the meaning of "substantial change": the age at which the limb end begins taking load on *every* curl.

5 Layer 4 — the replacement threshold

5.1 Criteria

An intervention is triggered at the earliest age at which any of three criteria reaches tolerance. The clearance criterion is modified so that contact occurs once the limb, pistoned in by u_{lift}, reaches the distal end:

$$a_c^*: \quad \varepsilon_c(a) = \varepsilon_{\max}, \tag{14}$$

$$a_d^*: \quad \delta(a) = u_{\text{lift}} \iff L(a) = D_s - u_{\text{lift}}, \tag{15}$$

$$a_\ell^*: \quad \Delta L(a) = \ell_{\text{fa}}(a) - \ell_{\text{fa}}(a_0) = \Delta L_{\max}, \tag{16}$$

$$\boxed{a^* = \min(a_c^*,\, a_d^*,\, a_\ell^*)}, \qquad \boxed{T^* = a^* - a_0}. \tag{17}$$

5.2 Analytic inversion

Tightness. Condition (14) with (10) gives the target circumference

$$C(a_c^*) = C_s(1 + \varepsilon_{\max}) = (1 + s_C)(1 + \varepsilon_{\max})\,C(a_0) \;=:\; C_{\text{tgt}}. \tag{18}$$

Insert (8) and solve. From $C(a) = C_{\text{tgt}}$,

$$1 + B_C e^{-r_C(a - a_b)} = \frac{K_C}{C_{\text{tgt}}} \;\Rightarrow\; e^{-r_C(a-a_b)} = \frac{1}{B_C}\left(\frac{K_C}{C_{\text{tgt}}} - 1\right), \tag{19}$$

hence

$$\boxed{a_c^* = a_b - \frac{1}{r_C}\ln\left[\frac{1}{B_C}\left(\frac{K_C}{C_{\text{tgt}}} - 1\right)\right]}, \qquad C_{\text{tgt}} = (1 + s_C)(1 + \varepsilon_{\max})\,C(a_0). \tag{20}$$

Clearance (with pistoning). Condition (15) gives the target length

$$L(a_d^*) = D_s - u_{\text{lift}} = L(a_0) + \delta_0 - u_{\text{lift}} \;=:\; L_{\text{tgt}}, \tag{21}$$

and the same inversion on (7) yields

$$\boxed{a_d^* = a_b - \frac{1}{r_L} \ln\left[\frac{1}{B_L}\left(\frac{K_L}{L_{\text{tgt}}} - 1\right)\right]}, \qquad L_{\text{tgt}} = L(a_0) + \delta_0 - u_{\text{lift}}. \tag{22}$$

The lift lowers L_{tgt} by u_{lift}, so a heavier load fires the clearance criterion at a younger age.

Length discrepancy. As the child grows, the intact arm lengthens while the prosthesis (built at a_0) does not; Ghidini identifies this prosthetic length discrepancy as a distinct replacement driver. Model the intact forearm length with the same logistic machinery,

$$\ell_{\text{fa}}(a) = \frac{K_\Lambda}{1 + B_\Lambda\, e^{-r_\Lambda (a - a_b)}}, \qquad B_\Lambda = \frac{K_\Lambda - \ell_{\text{fa}}(a_b)}{\ell_{\text{fa}}(a_b)}. \tag{23}$$

Condition (16) gives the target $\ell_{\text{fa}}(a_\ell^*) = \ell_{\text{fa}}(a_0) + \Delta L_{\max} =: \Lambda_{\text{tgt}}$, and the same inversion yields

$$\boxed{a_\ell^* = a_b - \frac{1}{r_\Lambda} \ln\left[\frac{1}{B_\Lambda}\left(\frac{K_\Lambda}{\Lambda_{\text{tgt}}} - 1\right)\right]}, \qquad \Lambda_{\text{tgt}} = \ell_{\text{fa}}(a_0) + \Delta L_{\max}. \tag{24}$$

Note $\ell_{\text{fa}}(a)$ is the *growing intact* forearm; its value *at fitting*, $\ell_{\text{fa}}(a_0)$, sets the fixed prosthesis moment arm in (12), so the two uses of forearm length are consistent and distinct.

Feasibility. The logarithms are real only if each limb can reach its target, and the socket must not bottom out at fitting:

$$C_{\text{tgt}} < K_C, \qquad L_{\text{tgt}} < K_L, \qquad \Lambda_{\text{tgt}} < K_\Lambda, \qquad \delta_0 > u_{\text{lift}}. \tag{25}$$

If a target exceeds its asymptote, that criterion never fires and is dropped from the min in (17) (the socket lasts to skeletal maturity on that axis).

6 Sensitivity analysis (the headline)

6.1 Monotonicity of the threshold in the design inputs

The growth curve $x(a)$ depends on θ only, not on the socket design. Differentiating the closed-form threshold ages shows the dependence on the design inputs is monotone and sign-definite. Using a_c^* from (20), the chain rule gives the compact result

$$\frac{\partial a_c^*}{\partial C_{\text{tgt}}} = \frac{K_C}{r_C\, C_{\text{tgt}}\,(K_C - C_{\text{tgt}})} \;> 0, \tag{26}$$

(obtained from $a_c^* = a_b - \frac{1}{r_C} \ln g$ with $g = \frac{1}{B_C}(\frac{K_C}{C_{\text{tgt}}} - 1)$, using $B_C g = \frac{K_C - C_{\text{tgt}}}{C_{\text{tgt}}}$). Since $C_{\text{tgt}} = (1 + s_C)(1 + \varepsilon_{\max})C(a_0)$,

$$\frac{\partial a_c^*}{\partial s_C} = \frac{K_C\,(1 + \varepsilon_{\max})\,C(a_0)}{r_C\, C_{\text{tgt}}(K_C - C_{\text{tgt}})} > 0, \qquad \frac{\partial a_c^*}{\partial \varepsilon_{\max}} = \frac{K_C\,(1 + s_C)\,C(a_0)}{r_C\, C_{\text{tgt}}(K_C - C_{\text{tgt}})} > 0. \tag{27}$$

Likewise, with $L_{\text{tgt}} = L(a_0) + \delta_0 - u_{\text{lift}}$,

$$\frac{\partial a_d^*}{\partial L_{\text{tgt}}} = \frac{K_L}{r_L\, L_{\text{tgt}}(K_L - L_{\text{tgt}})} > 0 \;\Rightarrow\; \underbrace{\frac{\partial a_d^*}{\partial \delta_0} > 0}_{\text{more clearance} \to \text{later}}, \qquad \underbrace{\frac{\partial a_d^*}{\partial u_{\text{lift}}} < 0}_{\text{heavier lift} \to \text{earlier}}. \tag{28}$$

Equation (28) is the quantitative statement that **more oversize or clearance always delays the threshold**, with no interior turning point: T^* is strictly increasing in s_C and δ_0, so the largest sensible values are set by the clinical ceiling beyond which a loose socket degrades control — a prescribed input, not something the model selects. The informative object is therefore the *sensitivity* of T^* to these inputs, reported as a surface over their ranges rather than a single recommended value.

6.2 Headline result: T^* versus fitting age

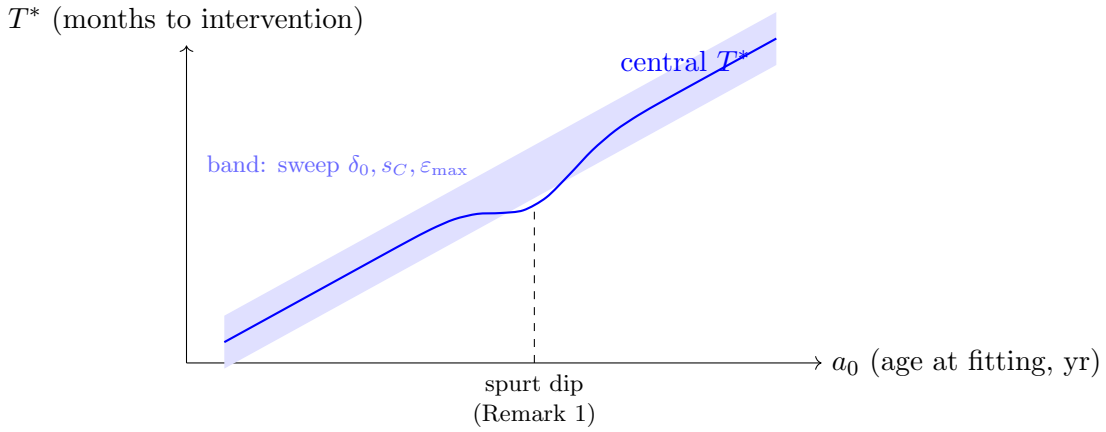
Combining (20)–(22),

$$T^*(a_0) = \min\big(a_c^*,\, a_d^*,\, a_\ell^*\big) - a_0, \tag{29}$$

is evaluated across fitting ages a_0, with a band obtained by sweeping the uncertain inputs δ_0, s_C, $\varepsilon_{\max}$ and $\Delta L_{\max}$ over literature ranges and u_{lift} over the lift scenarios. Its slope follows by the chain rule, e.g. through the tightness branch,

$$\frac{dT_c^*}{da_0} = \frac{\partial a_c^*}{\partial C_{\text{tgt}}}\,(1+s_C)(1+\varepsilon_{\max})\,f_C\big(C(a_0)\big) \;-\; 1, \qquad f_C(C) = r_C C(1 - C/K_C), \tag{30}$$

which grows toward late fitting (slower growth near the asymptote) until a criterion stops firing. A schematic of the headline figure:



(Schematic; axes qualitative. The real figure is produced once θ and the swept ranges are fixed.) Younger fitting ages give shorter service intervals; the curve answers the problem's actual question — whether and when a substantial change occurs — rather than prescribing a design.

6.3 Definition of "substantial change"

The point of substantial change is $a^* = \min(a_c^*, a_d^*, a_\ell^*)$: the earliest age at which the socket becomes intolerably tight ($\varepsilon_c = \varepsilon_{\max}$), the limb end begins bearing load on every curl ($\delta = u_{\text{lift}}$), or the intact-arm length discrepancy reaches tolerance ($\Delta L = \Delta L_{\max}$). $T^* = a^* - a_0$ is the predicted service interval.

7 Validation against clinical data

7.1 Proxy assumption (stated explicitly)

The available casefiles (Ghidini #1) are **lower-limb** data, whereas the model is for an **arm**. We therefore use them only as a proxy, and assume that only *dimensionless* quantities transfer between limbs: the tightness ratio ε_c, the tolerance $\varepsilon_{\max}$, and the residual-to-intact length ratio ρ. Absolute lengths and circumferences are not assumed to transfer.

7.2 Validate on the leg, predict on the arm

Legs and arms grow at different rates, so running the arm model and comparing it to lower-limb casefile intervals would test nothing. The honest procedure separates a structural test (on the limb for which data exist) from the prediction:

1. **Fit** the growth ODE (2) to lower-leg data (NHANES, or Ghidini #2 if access is granted), giving leg parameters θ_{leg}. The leg run uses its own, tighter tolerance $\Delta L_{\max}^{\text{leg}}$, since length discrepancy matters far more for walking than for an arm; being dimensional, $\Delta L_{\max}$ does not transfer between limbs (cf. § 7.1).

2. **Test the structure.** Run the full model with θ_{leg} across the observed fitting ages $a_{0,\text{obs}}$, producing a *distribution* of predicted intervals $\{T_i^*\}$. Compare it with the observed, right-censored Ghidini intervals $\{T_{\text{obs},i}\}$ (event indicator $e_i \in \{0,1\}$) through the Kaplan–Meier survivor estimate

$$\widehat{S}(t) = \prod_{t_i \le t} \left(1 - \frac{d_i}{n_i}\right), \tag{31}$$

   where d_i interventions occur among n_i sockets still in service at t_i. A match here tests the model's *mechanism*, not an accidental arm-vs-leg coincidence.

3. **Predict.** Swap in arm parameters θ_{arm} for the actual arm prediction.

Per-child fitting (θ_{child}) and a concordance measure are used only if the casefiles contain several measurements per child; otherwise the population-level comparison above stands.

8 Optional extension — compensatory musculoskeletal load

As a secondary output we add a *third action* — a straight-arm reach with the upper arm and forearm in line at $\varphi = 90°$ — and report a shoulder-torque overload index. With the shoulder as pivot, each mass acts at its own horizontal lever arm measured from the shoulder: the upper-arm length $\ell_{\text{ua}}(a)$ plus the fixed forearm arms ℓ_p, ℓ_h. Hence

$$\tau_{\text{sh}}(a) = g\big[m_p\big(\ell_{\text{ua}}(a) + \ell_p\big) + m_o\big(\ell_{\text{ua}}(a) + \ell_h\big)\big], \qquad S(a) \propto M_b(a)^{2/3}, \qquad \frac{\tau_{\text{sh}}(a)}{S(a)}. \tag{32}$$

Only $\ell_{\text{ua}}(a)$ grows — the prosthesis arms ℓ_p, ℓ_h are fixed at fitting — so the index is genuinely age-dependent: a rising value flags growth-driven overload at the shoulder. The 2/3 power is the standard mass-to-cross-section strength scaling. Reported, not swept, in the base model.

Summary of the pipeline

